Aim

To understand the features of **Wireshark** as a packet capture and network analysis tool and to study the encapsulation of data in network communication.

Theory

1. Wireshark

Wireshark is an open-source network protocol analyzer used to capture and examine packets that travel across a network in real time. It allows users to monitor network traffic and analyze different protocols operating at various layers of the **TCP/IP Model**.

Wireshark is widely used for network troubleshooting, protocol analysis, and learning how network communication works. It is commonly used by network administrators, cybersecurity professionals, and students studying computer networks.

Procedure

1. Install Wireshark on the system.
2. Open the Wireshark application.
3. Select the active network interface such as Wi-Fi or Ethernet.
4. Click on Start Capturing Packets.
5. Generate network traffic by opening a web browser and visiting a website.
6. After some packets are captured, stop the capture process.
7. Apply filters such as http, dns, or tcp to view specific packets.
8. Select any packet and expand the protocol layers to observe the encapsulation process.

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2. Features of Wireshark

Packet Capture
Wireshark can capture live network packets from different network interfaces such as Ethernet and Wi-Fi.

Deep Packet Inspection
It allows detailed inspection of network protocols like HTTP, DNS, TCP, UDP, and ARP.

Packet Filtering
Wireshark provides filtering options that help users focus on specific types of packets using capture filters and display filters.

Protocol Analysis
The tool automatically detects protocols and displays packet details according to their layers.

Color Coding
Different packets are displayed using various colors to make analysis easier.

Statistical Analysis
Wireshark provides statistical tools such as protocol hierarchy, packet graphs, and flow analysis.

Exporting Data
Captured packets can be saved and exported for further study or reporting.

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3. Packet Encapsulation

Encapsulation is the process of adding protocol information to data at each layer of the network model before transmission. Each layer of the TCP/IP model adds its own header information to the data.

The encapsulation process helps in proper delivery of data across the network.

Layer	Data Name	Added Information
Application	Data	User information
Transport	Segment	Port numbers (TCP/UDP header)
Network	Packet	IP addresses
Data Link	Frame	MAC addresses

Physical	Bits	Transmission of signals

Example of Encapsulation

When a user opens a website in a browser, the following process occurs:

- **Application Layer:** An HTTP request is created.
- **Transport Layer:** A TCP header containing source and destination port numbers is added.
- **Network Layer:** An IP header containing source and destination IP addresses is added.
- **Data Link Layer:** MAC addresses are added to form a frame.
- **Physical Layer:** The frame is converted into bits and transmitted over the network.

Wireshark captures these packets and shows the details of each layer, making it possible to observe how data is encapsulated during communication.

Result

The experiment was successfully performed using Wireshark. Network packets were captured and analyzed, and the encapsulation of data across different layers of the TCP/IP model was observed.